

Tide retrospective: from smooth tests to bounded tests

local total variation of Laplace families beyond all orders

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Abstract

If two smooth nonnegative losses have Laplace families that agree beyond all orders on every smooth compactly supported observable, then the local total variation of the two Boltzmann weights is beyond all orders: $\int \eta^2 |e^{-tL_2} - e^{-tL_1}| = o(t^{-\infty})$ for every smooth bump η , and $\int_K |e^{-tL_2} - e^{-tL_1}| = o(t^{-\infty})$ for every compact K . Consequently every bounded measurable compactly supported observable, in particular every continuous one, sees the two families agree beyond all orders, and normalized expectations of such observables agree once the partition values carry the anchor. This closes the observable-class gap left open by the day's two previous tides, whose transfer theorem was stated for the tested class C_c^∞ only. The mechanism is two pointwise inequalities, $D(a - b) \geq 0$ and $|a - b| \leq t|D|$ for $a = e^{-tL_1}$, $b = e^{-tL_2}$, $D = L_2 - L_1$, tested at the smooth observable $\eta^2 D$, and an AM–GM step with the free parameter t^{-N} that does the work of Cauchy–Schwarz with no square roots and no Hölder inequality. Six hundred and fifty lines, four LSP rounds, one whnf timeout bisected to a compact-support unification.

1 Setting

The projective-closure tide (2026-09-19-15-40) ended with a transfer theorem,

`superPoly_difference_of_projective.`

It says that projective agreement plus local equality near one zero gives exact agreement $I_2(\phi) - I_1(\phi) = o(t^{-\infty})$ for every ϕ in the tested class C_c^∞ . Its consult flagged the natural question, whether the conclusion extends to continuous observables, and sketched the answer: control the local total variation of the weights through the smooth observable $\eta^2 D$. This tide formalises that sketch. The deliberation (tide log 2026-09-19-17-00-tide-germbij-total-variation.md in tide-log/, consult in gpt_germbij_tv_v1.md) fixed the plan at the level of statements and settled the one design choice that mattered for Lean: replace the Cauchy–Schwarz step by an AM–GM bound with a free parameter, so that nothing about L^2 norms or square roots enters. Both parties voted for the full package.

2 The thing formalised

Write $a = e^{-tL_1}$, $b = e^{-tL_2}$, $D = L_2 - L_1$, and $I_i(\phi) = \int \phi e^{-tL_i}$.

Let $L_1, L_2 \geq 0$ be smooth and suppose $I_2(\phi) - I_1(\phi) = o(t^{-\infty})$ for every $\phi \in C_c^\infty$. Then:

- (V) $\int \eta^2 |b-a| = o(t^{-\infty})$ for every smooth compactly supported η , and $\int_K |b-a| = o(t^{-\infty})$ for every compact K .
- (U) $I_2(\phi) - I_1(\phi) = o(t^{-\infty})$ for every bounded measurable compactly supported ϕ , in particular for every continuous compactly supported ϕ .
- (N) If moreover $Z_1, Z_2 > 0$ carry polynomial lower bounds and $Z_2/Z_1 = 1 + o(t^{-\infty})$, then $I_2(\phi)/Z_2 - I_1(\phi)/Z_1 = o(t^{-\infty})$ for every such ϕ .

The Lean signature of (V):

```
theorem superPoly_integral_sq_mul_abs_exp_sub {L1 L2 : (iota -> R) -> R}
  (h1 : ContDiff R infty L1) (h2 : ContDiff R infty L2)
  (hL1 : forall w, 0 <= L1 w) (hL2 : forall w, 0 <= L2 w)
  (hexact : forall phi, ContDiff R infty phi -> HasCompactSupport phi ->
    SuperPoly fun t => (integral phi * exp (-(t * L2)))
      - integral phi * exp (-(t * L1)))
  {eta : (iota -> R) -> R} (heta : ContDiff R infty eta)
  (hetas : HasCompactSupport eta) :
  SuperPoly fun t =>
    integral (eta ^ 2 * |exp (-(t * L2)) - exp (-(t * L1))|)
```

(integrals abbreviated). The other declarations:

- the compact-set form, `superPoly_setIntegral_abs_exp_sub`;
- (U): `superPoly_difference_of_bounded` and `superPoly_difference_of_continuous`;
- (N): `superPoly_normalized_difference_of_bounded`;
- the closure wiring, `superPoly_difference_of_projective_bounded`;
- the pointwise lemmas `abs_exp_neg_sub_exp_neg_le`, `weightDiff_mul_nonneg` and `sq_exp_sub_le_mul`;
- the helper `superPoly_of_forall_eventually_le`;
- the helpers `SuperPoly.of_abs_le` and `SuperPoly.polyBounded_mul`.

Smoothness of both losses is used exactly once, to put $\eta^2 D$ in the tested class; the consult's remark that smoothness of D alone would do is recorded but not formalised. The conclusions are for compactly supported observables; nothing is claimed about tails, and the local L^∞ statement the consult showed to be true (via a gradient bound and interpolation) is left for a later tide.

3 Proof strategy

Pointwise. For $x, y \geq 0$, $|e^{-x} - e^{-y}| \leq |x - y|$: in the ordered case $e^{-x} - e^{-y} = e^{-x}(1 - e^{-(y-x)}) \leq y - x$ from $e^{-x} \leq 1$ and $1 - u \leq e^{-u}$ (`Real.add_one_le_exp`). The weight is

antitone, so $D(a - b) \geq 0$ for $t \geq 0$, and then $|a - b|^2 \leq |a - b|t|D| = tD(a - b)$.

Energy. $E(t) = \int \eta^2 D(a - b) = I_1(\eta^2 D) - I_2(\eta^2 D)$ is beyond all orders by hypothesis at the smooth observable $\eta^2 D$, and nonnegative pointwise.

AM–GM with a free parameter. For $\varepsilon = t^{-N}$, $|h| \leq \varepsilon/2 + h^2/(2\varepsilon)$, so $|a - b| \leq t^{-N}/2 + t^{N+1}D(a - b)/2$ and

$$\int \eta^2 |a - b| \leq \frac{1}{2} \left(\int \eta^2 \right) t^{-N} + \frac{1}{2} t^{N+1} E(t) \leq C_N t^{-N}$$

eventually, using $E(t) \leq t^{-(2N+1)}$. Since this holds for every N , the integral is beyond all orders (`superPoly_of_forall_eventually_le`, which turns an $O(t^{-N})$ bound for all N into $o(t^{-N})$ for all N by shifting the index).

Bounded tests. A `ContDiffBump` at the origin equal to 1 on a closed ball containing the support of ϕ gives $\eta^2 = 1$ on $\text{supp } \phi$, so $|I_2(\phi) - I_1(\phi)| \leq \int |\phi| |b - a| \leq M \int \eta^2 |b - a|$. Integrability of ϕe^{-tL_i} for $t \geq 0$ comes from a small lemma, `integrable_of_bounded_of_hasCompactSupport`, via domination by the indicator of the support. Continuous ϕ is bounded by `HasCompactSupport.exists_bound_of_continuous`.

Normalized tests. $I_2/Z_2 - I_1/Z_1 = (I_2 - I_1)/Z_1 + (I_2/Z_2)(1 - Z_2/Z_1)$; the first term is (U) times a polynomially bounded $1/Z_1$, the second is $Z_2/Z_1 - 1$ times a polynomially bounded factor ($|I_2| \leq M \text{vol}(\text{supp } \phi)$ and $1/Z_2 = O(t^n)$), both by `SuperPoly.polyBounded_mul`.

4 Roadblocks and resolutions

Four LSP rounds. Three were ordinary: an `isLittle0_one_iff` that needs its normed type passed explicitly; an absolute-value orientation; a parenthesisation slip where an integral written as `integral w, eta w ^ 2 + ...` swallowed the rest of the line into the integrand (the pretty-printer shows the same text either way, so read integrals with a following `+` twice); a sign lemma I had first stated without $t \geq 0$, which is false for $t < 0$ and which `linarith` rightly refused; and the `Measure.real` versus `toReal` spelling of `setIntegral.const`.

The fourth was a deterministic whnf timeout attributed to the whole compact-set lemma, with every `have` checking. Bisection by inserting `sorry` at successive points located it to one term,

```
((heta.continuous.pow 2).mul hc).integrable_of_hasCompactSupport
(heta2s.mul_right)
```

where `mul_right` produces `HasCompactSupport (f * g)` in `Pi` form and the expected type is a `lambda`; the unifier tried to unfold its way through. Naming the compact-support fact first with the `lambda` type ascribed (`have hsupp : HasCompactSupport fun w => ...` followed by `:= heta2s.mul_right`) and passing `hsupp` elaborated instantly. The identical construction had worked two lemmas earlier by luck of the elaboration order. This is the `Pi.add` gotcha of `CLAUDE.md` in a new coat.

The consult, before writing, settled the design and scoped the claims. On the AM–GM route:

GPT-6 Astra: The AM–GM route loses nothing at the SuperPoly level. [...] The one bookkeeping point: to conclude little- o at exponent N , use the big- O estimate at exponent $N + 1$. Your proposed helper is exactly right.

On regularity:

GPT-6 Astra: Both losses being smooth is sufficient, but not genuinely necessary. What this proof needs for the test is $\eta^2(L_2 - L_1) \in C_c^\infty$. [...] I would not promote that into a necessity theorem.

And on the scope of the normalized corollary:

GPT-6 Astra: (U) covers bounded measurable compactly supported φ . It does not establish this normalized conclusion for arbitrary globally bounded measurable tests. Partition-function agreement alone does not control total variation in remote tails.

5 What was Mathlib, what was new

Mathlib, all lookup:

- exponentials: `Real.add_one_le_exp`, `Real.exp_le_exp`, `Real.exp_le_one_iff`;
- asymptotics: `isLittle0_one_iff` and `tendsto_inv_atTop_zero`;
- big- O algebra: `IsBig0.trans_isLittle0`;
- integrals: `integral_mono`, `integral_add`, and `integral_const_mul`;
- set integrals: `setIntegral_congr_fun` and `setIntegral_const`;
- comparison: `setIntegral_le_integral`;
- integrability: `Integrable.mono'`, `integrable_indicator_iff`, `integrableOn_const`, `IsCompact.measure_lt_top`;
- supports: `HasCompactSupport.comp_left`;
- bounds: `HasCompactSupport.exists_bound_of_continuous`;
- balls: `Metric.isBounded_iff_subset_closedBall`.

Seabed: `SuperPoly` and its closure lemmas, `integrable_mul_exp_neg_of_compactSupport`, the `gapBump` helper and `ProjectiveAgreement` from the closure tide, and the transfer theorem it wires into.

New: the three pointwise inequalities, three `SuperPoly` helpers, the bounded-compact-support integrability lemma, the smooth bump on a compact set, and the five theorems. About 650 lines.

6 Lessons

- For `CLAUDE.md`: a `whnf` timeout attributed to a whole declaration is bisected fastest by inserting `sorry` at successive tactic positions and re-checking; three runs located this one. And compact-support facts built by `mul_right` should be named with their `lambda` type before use.
- For the tide skill: when the consult of the previous tide names a follow-up with a proof sketch, the next tide can go straight to a statement-level consult; this one needed one round and no revision.
- For the paper: the observable class of the closure theorems is now C_c^0 , indeed bounded measurable with compact support. The normalized/unnormalized distinction of the note collapses, modulo $o(t^{-\infty})$, on that class as soon as one common zero anchors the gauge.

7 Follow-ups

- *Local L^∞* . Under smoothness, $\|\nabla(a - b)\| \leq Ct$ on a fixed ball, so $\sup_K |a - b| \leq C_K (t^d \int_{K'} |a - b|)^{1/(d+1)}$: local total variation beyond all orders implies local uniform agreement beyond all orders. Needs a Lipschitz-peak interpolation argument.
- *Smooth difference only*. Restate (V) with L_1 merely continuous and $D = L_2 - L_1$ smooth.
- *t -dependent tests*. The compact-set form is uniform over $|\phi_t| \leq M$ supported in K ; state it.
- *Paper*. Fold (U) into the closure theorem's statement as the final observable class; one sentence on the AM–GM mechanism.

Acknowledgements

GPT-6 Astra proposed the mechanism in the previous tide's consult, confirmed the AM–GM replacement of Cauchy–Schwarz, and scoped the normalized corollary. The seabed is the day's three germbij tides.